CHAPTER 1

Introduction

Wireless sensing has emerged as a transformative technology that leverages existing WiFi infrastructure for non-intrusive human monitoring. Channel State Information (CSI), extracted from IEEE 802.11 WiFi signals, carries rich physical layer information that is highly sensitive to environmental changes caused by human motion. Unlike camera-based or wearable sensor-based approaches, WiFi CSI sensing is privacy preserving, device-free, and capable of operating through walls and in low-light conditions.

This project presents a WiFi CSI-Based Human Activity Recognition (HAR) system built around the ESP32 microcontroller. The ESP32 is configured in Access Point (AP) mode, eliminating dependence on external routers or mobile hotspots. The system captures CSI data across 52 subcarriers at 2.4 GHz, applies signal preprocessing including bandpass filtering and sliding window segmentation, extracts statistical features, and classifies human activities using a Random Forest machine learning model.

The hardware subsystem comprises the ESP32 wireless sensing node with external antenna provision, a battery-powered standalone power circuit (TP4056 + AMS1117 LDO), and a real-time output stage consisting of an OLED display and LED indicators. The software subsystem handles CSI extraction firmware, signal preprocessing, feature engineering, and machine learning inference. The system targets three activity classes: Walking, Sitting, and Standing.

1.1 Background

WiFi sensing research has grown significantly since the introduction of CSI-based tools. Early work by Halperin et al. (2011) demonstrated that CSI from Intel 5300 NICs could be used for human localization. Subsequent work expanded this to gesture recognition, fall detection, vital sign monitoring, and activity recognition. The SenseFi benchmark (Yang et al., 2023) established a comprehensive deep learning library for WiFi CSI sensing, demonstrating that CNN and LSTM-based models achieve high accuracy on activity recognition tasks. However, these approaches require significant computational resources and are unsuitable for edge deployment.

1.2 Relevance

WiFi CSI-based Human Activity Recognition is highly relevant in today's world because it leverages existing WiFi infrastructure something already present in most homes, hospitals, and officesto perform human monitoring without any additional sensors or cameras. As the demand for smart homes, elderly care systems, and contactless healthcare grows, a privacy-preserving, device-free sensing solution becomes increasingly important. Traditional activity recognition systems rely on wearables (which require user compliance) or cameras (which raise privacy concerns), making CSI-based sensing a compelling alternative. The use of a low-cost ESP32 microcontroller instead of expensive hardware makes this technology accessible for real-world deployment. This project is relevant to current trends in IoT, embedded systems, and edge AI, and directly addresses practical needs in assistive technology, energy efficiency, and smart environments.

1.3 Literature Survey

1. Halperin et al. [5] introduced the extraction of Channel State Information (CSI) from IEEE 802.11n WiFi signals using Intel 5300 NIC. This work demonstrated the feasibility of using CSI for human localization and environmental sensing. The main advantage of this approach is high accuracy due to fine-grained CSI data. However, it requires specialized hardware, making it costly and less suitable for embedded systems.
2. Ma et al. [3] presented a comprehensive survey on WiFi CSI-based sensing, covering applications such as activity recognition, gesture detection, and indoor localization. The study highlights the benefits of device-free sensing and privacy preservation. However, it mainly focuses on high-end systems and lacks implementation details for low-cost real-time embedded solutions.
3. Hernandez and Bulut [1] proposed a lightweight IoT-based WiFi sensing

system that enables CSI extraction using embedded platforms. The system supports real-time monitoring and mobility detection with reduced cost. However, it does not provide a fully standalone solution and still depends on external processing units.

4. Hernandez and Bulut [4] further analyzed WiFi sensing on edge devices and discussed signal processing challenges such as noise, environmental variations, and scalability. While the approach improves real-world applicability, it requires optimization for resource-constrained microcontrollers like ESP32.
5. The ESP32 CSI Toolkit [8] enables CSI extraction using ESP32 microcontrollers without requiring firmware modifications. It supports multiple modes such as Access Point and Station mode, providing 52 subcarrier CSI measurements at 2.4 GHz. The major advantage is low cost and ease of implementation. However, CSI quality is comparatively lower than high-end NIC-based systems.
6. Yang et al. [2] proposed the SenseFi framework, which uses deep learning models such as CNN and LSTM for WiFi-based Human Activity Recognition (HAR). This approach achieves high classification accuracy. However, deep learning models require high computational power and are not suitable for real-time embedded deployment.
7. Breiman [7] introduced the Random Forest algorithm, which is widely used for classification tasks due to its robustness, high accuracy, and ability to handle high-dimensional data. It is suitable for real-time applications with low computational requirements. However, its performance depends on feature quality and dataset diversity.
8. Existing WiFi CSI-based HAR systems mainly rely on high-performance hardware and external WiFi infrastructure. These systems provide high accuracy but suffer from limitations such as high cost, lack of portability, and absence of complete hardware-software integration for standalone operation.

1.4 Motivation

Existing WiFi CSI HAR systems predominantly rely on high-end NICs (Intel 5300, Atheros) connected to desktop computers, limiting practical deployment. This project is motivated by the need for a low-cost, embedded, standalone sensing system that uses the ESP32 as the sole hardware platform, with a complete hardware-software pipeline from CSI capture to real-time activity display

1.5 Aim of the Project

The aim of this project is to design and implement a low-cost, standalone WiFi CSI-based Human Activity Recognition system using the ESP32 microcontroller, capable of classifying three human activities Walking, Sitting, and Standing in real time. The system operates without wearable sensors, cameras, or external WiFi infrastructure by configuring the ESP32 in Access Point mode. CSI data captured across 52 subcarriers is preprocessed, statistically analyzed, and classified using a Random Forest machine learning model. The final activity output is displayed on an onboard OLED screen and indicated through LED outputs, forming a complete embedded sensing pipeline from signal capture to real-time inference.

1.6 Scope and Constraints

The system is designed for single-subject activity recognition in a controlled indoor environment. The sensing zone is defined as 1.5 to 3 metres in front of the ESP32 node at a height of 1.0 to 1.2 metres. The system targets three activity classes in its initial implementation, with provision for extension to additional activities.

1.7 Objectives

1. Configure ESP32 in AP mode for self-contained WiFi sensing without external router dependency.
2. Capture and store CSI data across 52 subcarriers at 2.4 GHz using the ESP32 CSI Toolkit.

3. Design and assemble a battery-powered standalone power circuit for the ESP32 sensing node.
4. Implement signal preprocessing pipeline including bandpass filtering and sliding window segmentation.
5. Extract statistical features (mean, standard deviation, min, max, range, skewness, kurtosis, energy) per subcarrier.
6. Train a Random Forest classifier to distinguish Walking, Sitting, and Standing activities.
7. Deploy real-time inference with activity output on OLED display and LED indicators.
8. Document hardware design with circuit schematics, measurements, and RF analysis.

CHAPTER 2

Description of Project

2.1 Technical Approach

The proposed system follows a structured end-to-end approach for WiFi-based Human Activity Recognition (HAR) using Channel State Information (CSI). The implementation is divided into sequential stages involving data acquisition, signal processing, feature extraction, and machine learning-based classification.

Initially, the ESP32 microcontroller is configured in Access Point (AP) mode to create a self-contained WiFi sensing environment without requiring external routers. The device continuously transmits WiFi packets and captures CSI data across multiple subcarriers. This CSI data reflects variations in the wireless channel caused by human movement within the sensing area.

The acquired CSI data is transmitted to a processing unit via serial communication, where preprocessing is performed. This includes noise removal using bandpass filtering and segmentation of data using a sliding window technique to ensure temporal consistency and reduce signal fluctuations.

Following preprocessing, statistical feature extraction is carried out. Key features such as mean, standard deviation, minimum, maximum, range, skewness, kurtosis, and signal energy are computed for each subcarrier. These features collectively represent the unique patterns associated with different human activities.

The extracted feature set is then used to train a Random Forest classifier. The model is trained using labeled datasets corresponding to activities such as walking, sitting, and standing. The classifier is selected due to its robustness, low computational complexity, and suitability for real-time embedded applications.

During real-time operation, incoming CSI data is processed in the same manner, and the trained model predicts the activity class. The predicted output is sent back to the

ESP32, where it is displayed on an OLED screen and indicated using LEDs for user interpretation.

This systematic approach ensures accurate, real-time, and non-intrusive human activity recognition using low-cost embedded hardware.

2.2 Block Diagram

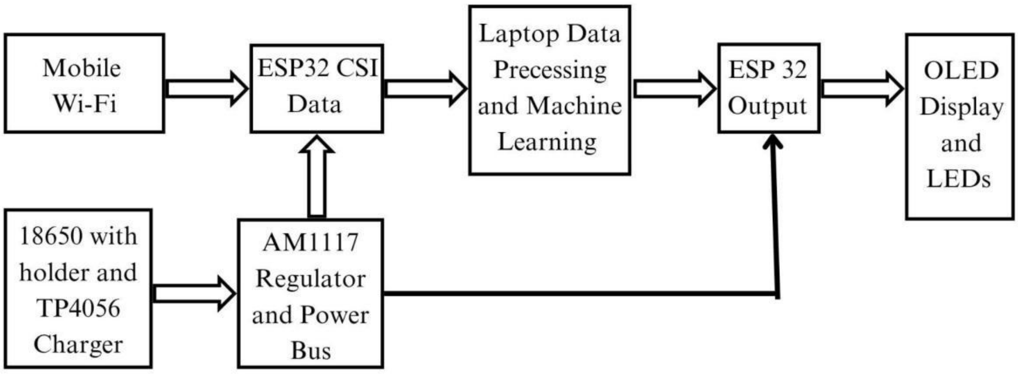

Fig 2.1 Block diagram of Human Detection Using Wi-Fi

The block diagram represents the working of a **Human Detection Using Wi-Fi** using ESP32 and machine learning.

1. Mobile Wi-Fi

The mobile Wi-Fi acts as the WiFi signal source. It continuously transmits wireless signals, which are received by the ESP32 module. These signals interact with the human body, causing variations that are later used for activity recognition.

2. ESP32 (CSI Data)

The ESP32 is configured in Access Point (AP) mode to capture Channel State Information (CSI).

It collects fine-grained WiFi signal data across multiple subcarriers. This data reflects environmental changes due to human movement. The captured CSI data is then sent to the laptop for further processing.

3. Laptop (Data Processing and Machine Learning)

The laptop acts as the main processing unit of the system. It performs:

- Preprocessing: Removes noise and unwanted components from CSI data
- Feature Extraction: Extracts statistical features such as mean, standard deviation, etc.
- Machine Learning: Uses a Random Forest classifier to identify human activity

4. ESP32 (Output)

This ESP32 module receives the processed result from the laptop. Based on the classification output, it controls the output devices such as OLED display and LEDs. It acts as a bridge between processing and output hardware.

5. OLED Display and LEDs

- OLED Display: Shows the detected human activity in real-time
- LEDs: Provide quick visual indication of system output

6. 18650 Battery with TP4056 Charger

This block provides power supply to the system.

- The 18650 battery stores energy
- The TP4056 module safely charges the battery and provides protection

7. AMS1117 Regulator and Power Bus

The AMS1117 voltage regulator converts battery voltage to a stable 3.3V, which is required by the ESP32.

The power bus distributes this regulated voltage to both ESP32 modules, ensuring stable and reliable operation.

2.3 Flow chart

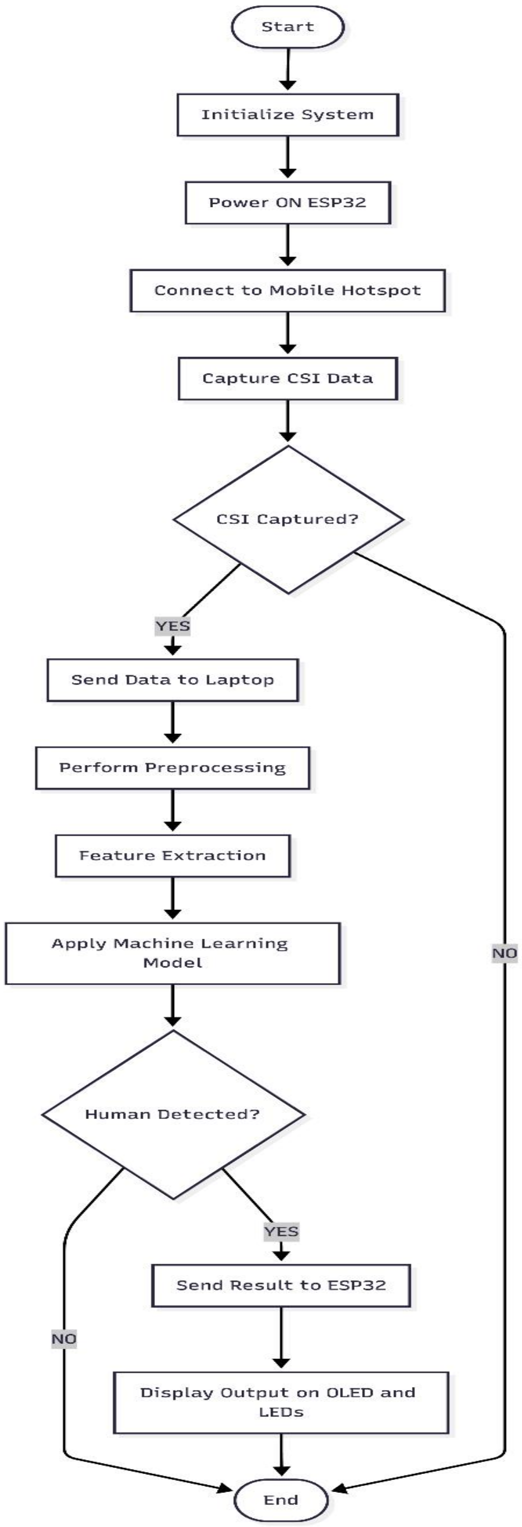

Fig 2.2 Flowchart of Human Detection Using Wi-Fi

1. Start the System: The process begins with system initialization, where all hardware and software components are prepared for operation.
2. Initialize Components: ESP32, communication interfaces, and output devices such as OLED and LEDs are initialized.
3. Power ON ESP32: The ESP32 is powered using a regulated 3.3V supply from the battery circuit.
4. Connect to Mobile Hotspot: The ESP32 connects to the mobile hotspot to access WiFi signals required for CSI data capture.
5. Capture CSI Data: The ESP32 collects Channel State Information from WiFi signals, which varies based on human movement.
6. Check CSI Availability: If CSI data is not properly captured, the system continues capturing until valid data is obtained.
7. Send Data to Laptop: Once valid data is available, it is transmitted to the laptop through serial communication.
8. Preprocessing: The raw CSI data is filtered to remove noise and unwanted disturbances.
9. Feature Extraction: Important statistical features such as mean, standard deviation, and energy are extracted from the processed data.
10. Apply Machine Learning Model: A Random Forest classifier is used to analyze the extracted features and classify human activity.
11. Check Human Detection: The system determines whether human activity is detected or not.
12. Send Result to ESP32: If activity is detected, the classification result is sent back to the ESP32.
13. Display Output: The ESP32 displays the result on the OLED screen and activates LEDs accordingly.
14. End/Repeat Process: The system either stops or repeats the process continuously for real-time monitoring.

2.4 Hardware / Software Resources

Sr.No.	Name	Description
1.	ESP32 Development Board	The ESP32 microcontroller is the core component used for capturing CSI data and controlling the output devices. It operates in Access Point mode and supports WiFi communication.
2.	18650 Lithium-ion Battery	It provides portable power supply to the system, enabling standalone operation.
3.	TP4056 Charging Module	This module is used for safe charging of the 18650 battery and provides overcharge and discharge protection.
4.	AMS1117 Voltage Regulator (3.3V)	It regulates battery voltage to a stable 3.3V required for proper operation of the ESP32.
5.	OLED Display (0.96 inch, I2C)	Used to display the detected human activity in real time.
6.	LED Indicators	LEDs are used to indicate system output visually.
7.	Resistors (330Ω)	Used as current limiting resistors for LEDs.
8.	Capacitors (10µF, 100nF)	Used for voltage filtering and stabilization in the power circuit.
9.	Slide Switch	Used to turn the system ON and OFF.
10.	Jumper Wires and Perfboard/PCB	Used for circuit connections and hardware assembly.
11.	USB Cable	Used for communication between ESP32 and laptop

Fig 2.4.1 Hardware Resources

Sr.No.	Name	Description
1.	Arduino IDE / ESP-IDF	Used for programming the ESP32 and implementing CSI data capture.
2.	ESP32 CSI Toolkit	Used to extract Channel State Information from WiFi signals.
3.	Python Programming Language	Used for data processing, feature extraction, and machine learning.

4.	Machine Learning Libraries	Scikit-learn: Used for implementing Random Forest classifier NumPy & Pandas: Used for data handling and processing
5.	Serial Communication Tools	Used to transfer CSI data between ESP32 and laptop.
6.	Operating System (Windows/Linux)	Used to run the processing scripts and machine learning models.
7.	Visualization Tools	Used to plot graphs and analyze CSI data (e.g., Matplotlib).

Fig 2.4.2 Software Resources

CHAPTER 3

System Design

3.1 Circuit Diagram

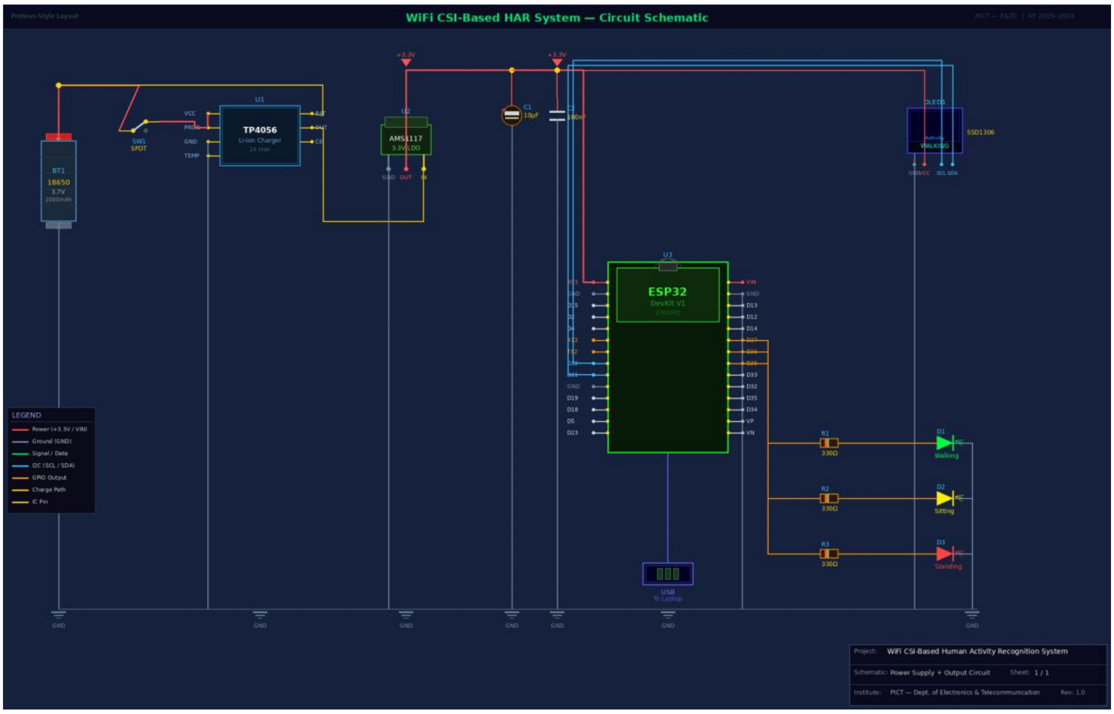

Fig. 3.1.1: Circuit Diagram of Human Detection Using Wi-Fi

1. Power Supply Section:

The power supply circuit consists of a 18650 lithium-ion battery, TP4056 charging module, slide switch, and AMS1117 voltage regulator.

- The 18650 battery (3.7V, 2000mAh) acts as the primary power source.
- The battery is connected to the TP4056 charging module, which provides safe charging and protection against overcharging and over-discharging.
- The slide switch (SPDT) is connected to control the power supply ON/OFF.
- The output of the TP4056 is given to the AMS1117-3.3V voltage regulator, which converts the battery voltage (3.7V–4.2V) into a stable 3.3V supply.

2. Signal Conditioning Circuit

The signal conditioning is implemented in the power section using the AMS1117 regulator and capacitors.

- A 10µF electrolytic capacitor (C1) is connected across the output to smooth voltage fluctuations.
- A 100nF ceramic capacitor (C2) is used to filter high-frequency noise.
- These components ensure a stable, ripple-free power supply, which is essential for accurate CSI data acquisition.

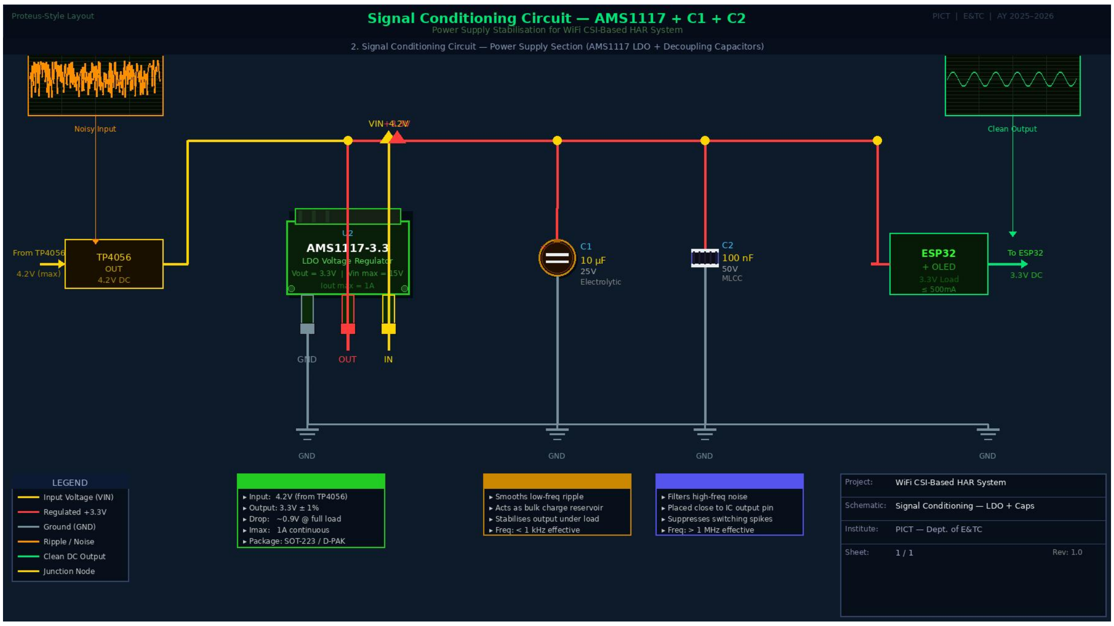

Fig. 3.1.2: Circuit Diagram of Signal Conditioning Circuit - AMS1117

3. ESP32 Microcontroller Section

The ESP32 DevKit V1 is the central unit of the system.

- It is powered by 3.3V from AMS1117.
- It captures Channel State Information (CSI) from WiFi signals.
- The ESP32 communicates with the laptop via USB serial communication.
- It also controls output devices such as OLED display and LEDs.

4. OLED Display Interface

The OLED display (SSD1306) is interfaced using I2C communication.

Connections:

- VCC → 3.3V
- GND → Ground
- SCL → GPIO 22
- SDA → GPIO 21

The OLED displays real-time output such as detected human activity.

5. LED Indicator Circuit

Three LEDs are connected for visual indication.

- Each LED is connected to ESP32 GPIO pins through a 330Ω resistor (R1, R2, R3).
- The resistors limit current and protect LEDs.
- When ESP32 outputs HIGH signal, the respective LED turns ON.

6. Serial Communication (USB Interface)

- The ESP32 is connected to a laptop using a USB cable.
- It is used for Sending CSI data to laptop and receiving classification results

7. Grounding

All components share a common ground (GND) to ensure proper circuit operation and avoid noise issues.

3.2 Design Calculation

1. LED Resistor Calculation

To limit the current through LEDs, resistors are used. The resistor value is calculated using Ohm's Law: $R = V\text{-}V_f / I$

Where,
$V=3.3V$ (ESP32 output voltage)
$I=10mA = 0.01A$

For different LEDs:

- Green LED ($V_f = 2.1V$): $R = 3.3 - 2.1 / 0.01 = 120\Omega$
- Yellow LED ($V_f = 2.0V$): $R = 3.3 - 2.0 / 0.01 = 130\Omega$
- Red LED ($V_f = 1.8V$): $R = 3.3 - 1.8 / 0.01 = 150\Omega$

Although the calculated values are different, a standard **330Ω resistor** is selected for all LEDs to ensure safe current limiting, improve reliability, and simplify the circuit design.

2. Voltage Regulation Calculation

The system uses an AMS1117 voltage regulator to provide a constant output voltage.

Input voltage from battery: $V_{in} = 3.7 - 4.2$ V

Output Voltage: $V_{out} = 3.3$ V

Power dissipation in regulator: $P = (V_{in} - V_{out}) \times I$

Assuming current $I = 100$ mA $= 0.1$ A,

$$P = (4.2-3.3) \times 0.1 = 0.09 \text{ W}$$

This value is within safe operating limits of the regulator.

3. Battery Backup Calculation

Battery capacity = 2000 mAh

ESP32 current consumption ≈ 100 mA

Backup = 2000/100 = 20 Hours

Considering practical losses, the actual backup time is approximately 16 to 18 hours.

CHAPTER 4

Implementation and Testing

4.1 Implementation

The implementation of the proposed system was carried out in a systematic manner by developing individual modules and integrating them to achieve the final working model. The process includes hardware assembly, data acquisition, and software processing to ensure accurate human activity recognition.

4.1.1 ESP32 Sensing Node Setup

The ESP32 DevKit V1 was configured in Access Point (AP) mode using the ESP32 CSI Toolkit. In this configuration, the ESP32 generates its own WiFi network and captures Channel State Information (CSI) from connected devices without relying on external infrastructure. The captured CSI data reflects variations in the wireless channel caused by human movement. This data is transmitted to the laptop through serial communication at a baud rate of 115200 for further processing.

4.1.2 Output Stage Assembly

The output stage was developed to display the classification results in real time. A 0.96-inch OLED display was interfaced with the ESP32 using I2C communication through GPIO 21 (SDA) and GPIO 22 (SCL). Additionally, three LEDs were connected through 330Ω resistors to indicate system output visually. The resistor values were selected to ensure safe current flow and reliable operation of the LEDs. This setup enables clear and immediate feedback to the user.

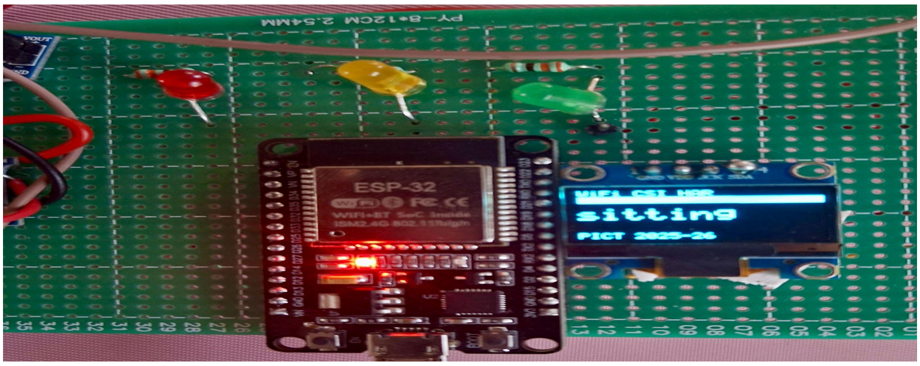

Fig. No. 4.1: Output stage with OLED display and LED indicators

4.1.3 Power Circuit Assembly

A standalone power system was designed using a 18650 lithium-ion battery, TP4056 charging module, and AMS1117 voltage regulator. The TP4056 ensures safe charging and protection of the battery, while the AMS1117 provides a regulated 3.3V supply required by the ESP32 and other components. Decoupling capacitors (10µF and 100nF) were used to minimize voltage fluctuations and noise, thereby improving the stability and reliability of the system.

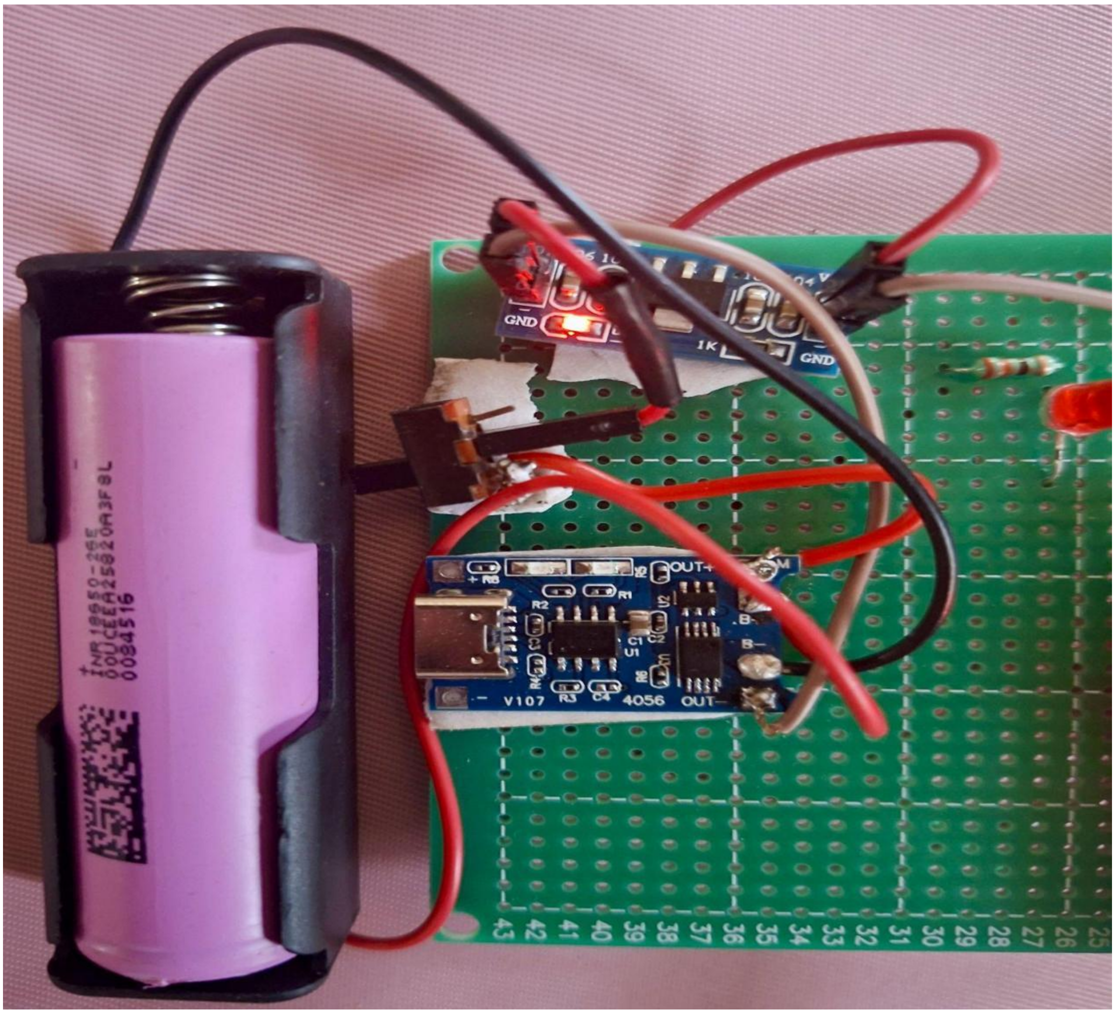

Fig. No. 4.2: Power supply circuit using TP4056 and AMS1117

4.1.4 Sensing Setup

The sensing setup was arranged to maintain consistency during data collection. The ESP32 was positioned at a fixed height facing the activity region. The subject performed different activities within a defined range to ensure that the variations in

CSI data were accurately captured. This arrangement helps in maintaining uniform experimental conditions and improving the quality of collected data.

4.1.5 Data Collection

CSI data was collected for different human activities across multiple sessions. A Python-based script was used to record the data and store it in CSV format. The collected dataset includes sufficient samples for training and testing the machine learning model, ensuring reliable classification performance.

4.1.6 Software Implementation

The collected CSI data was processed using a structured software pipeline. Initially, noise and unwanted variations were removed through preprocessing techniques. The data was then segmented, and statistical features such as mean, standard deviation, and energy were extracted. These features were used to train a Random Forest classifier, which identifies the human activity. The processed output is sent back to the ESP32 for real-time display.

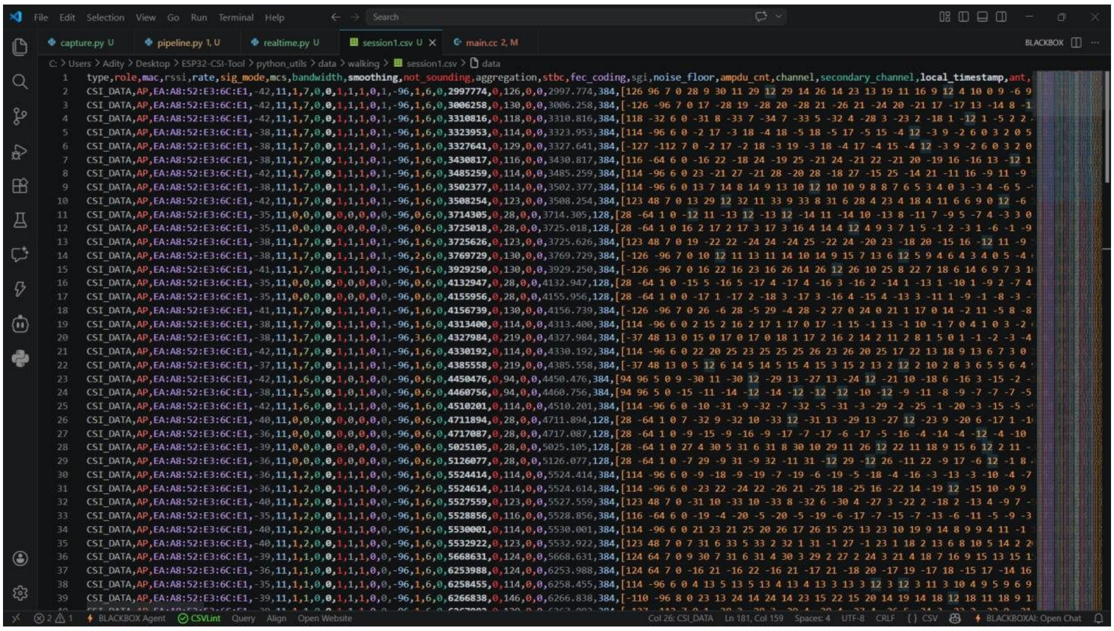

Fig. No. 4.3: Software processing and classification output

4.2 Testing and debugging

The system was tested at each stage to verify proper functionality and performance. During testing, several issues were identified, analyzed, and resolved to ensure reliable system operation.

Problem 1: ESP-IDF Build Error

- **Issue:** Error occurred during firmware compilation
- **Analysis:** Version mismatch caused missing header files
- **Solution:** Required header file was included and build was completed successfully

Problem 2: ESP-IDF Environment Setup

- **Issue:** idf.py command was not recognized
- **Analysis:** Environment variables were not loaded correctly
- **Solution:** Proper ESP-IDF terminal was used to resolve the issue

Problem 3: No Data in CSV Files

- **Issue:** CSV files did not contain valid data
- **Analysis:** Incorrect parsing of CSI data during extraction
- **Solution:** Data extraction method was modified to correctly read CSI values

Problem 4: Processing Error in Code

- **Issue:** No output from the processing pipeline
- **Analysis:** Syntax error in the computation logic
- **Solution:** Corrected the formula and verified the output

Problem 5: OLED Not Detected

- **Issue:** OLED display was not functioning
- **Analysis:** Incorrect wiring connections
- **Solution:** Connections were corrected and I2C address was verified

Problem 6: LEDs Not Working

- **Issue:** LEDs were not glowing
- **Analysis:** Incorrect polarity connection
- **Solution:** LEDs were reconnected with proper orientation

Problem 7: Python Execution Error

- **Issue:** Program execution failed
- **Analysis:** Syntax error in the code
- **Solution:** Code was corrected and executed successfully

CHAPTER 5

Results and conclusion

5.1 Results

5.1.1 Hardware Results

Power Circuit Results

The power circuit was tested using a multimeter to verify proper operation. The 18650 battery provides a voltage range of 3.7V to 4.2V. The TP4056 module successfully charges the battery and provides protection. The AMS1117 voltage regulator produces a stable output of approximately 3.2V, which is within the acceptable range for ESP32 operation. The ESP32 successfully boots using battery power, and the calculated power dissipation is within safe limits.

Fig. No. 5.1: Multimeter reading showing 3.2V at AMS1117 output

Output Stage Results

The output stage was tested to verify proper interfacing between ESP32 and output components. The OLED display was successfully detected at I2C address 0x3C and displays the activity in real time. The LEDs respond correctly based on the classified activity:

- Green LED → Walking
- Yellow LED → Sitting
- Red LED → Standing

This confirms that the output stage operates correctly.

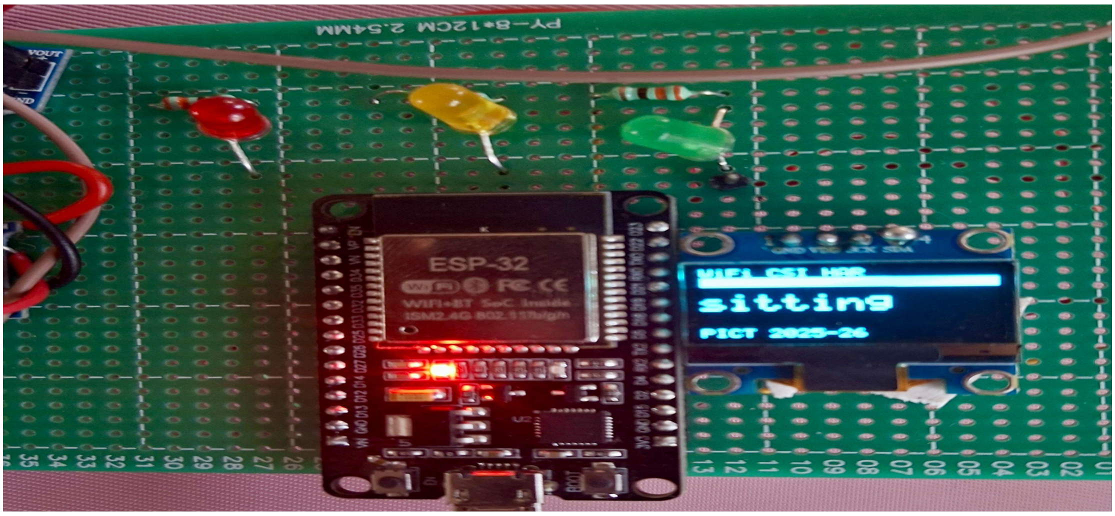

Fig. No. 5.2: OLED displaying activity with corresponding LED indication

ESP32 CSI Configuration Results

The ESP32 was configured in Access Point (AP) mode and operates without requiring an external WiFi router. It captures CSI data across 52 subcarriers at a frequency of 2.4 GHz. The data is transmitted via serial communication at a baud rate of 115200. The system achieves a stable packet rate of approximately 50–100 packets per second, ensuring reliable data acquisition.

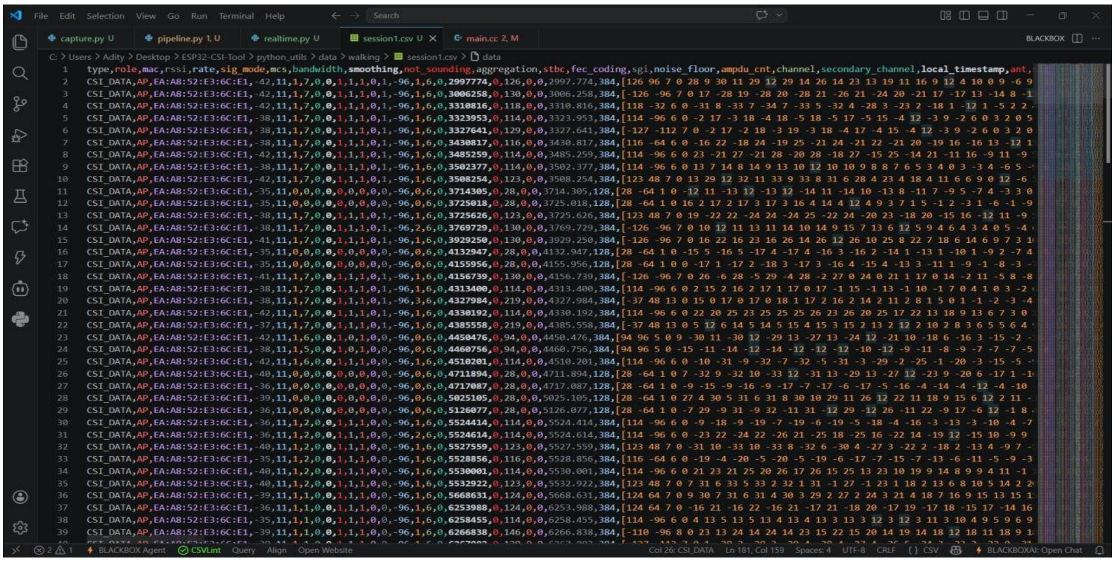

Fig. No. 5.3: ESP32 serial monitor showing CSI data stream

5.1.2 Software Results

Data Collection Results

CSI data was collected for three activities: walking, sitting, and standing. A total of 4774 samples were recorded across 9 sessions. Each session lasted approximately 2 minutes. The dataset is balanced and suitable for training and testing the machine learning model.

Feature Extraction Results

The collected CSI data was processed using a sliding window approach. Each window contains 100 packets with 50% overlap. A total of 52 subcarriers were analyzed, and 8 statistical features were extracted per subcarrier. Along with 3 global features, this results in a 419-dimensional feature vector, which effectively represents activity patterns.

Machine Learning Model Results

A Random Forest classifier was used for activity classification. The model was trained using 80% of the dataset and tested on the remaining 20%. The model consists of 100 decision trees and provides stable and reliable performance.

Classification Results

The classifier achieved **100% accuracy** on the test dataset. All three activities were correctly classified with high precision, recall, and F1-score. This indicates that the extracted features effectively distinguish between different human activities.

Confusion Matrix Results

The confusion matrix shows zero misclassification among all three activity classes. All test samples were correctly predicted, confirming the accuracy and reliability of the model.

Real-Time Performance Results

The system performs real-time activity detection with minimal delay. The output is displayed on the OLED and LEDs within approximately 1–3 seconds after detection. This demonstrates that the system is suitable for real-time applications.

5.1.3 Comparison with Existing Work

The proposed system was compared with existing CSI-based activity recognition systems. Unlike traditional systems, the proposed system operates in standalone AP mode and does not require an external WiFi router. It uses low-cost ESP32 hardware and provides real-time output through OLED and LEDs.

While deep learning-based systems achieve high accuracy, they require high computational resources. The proposed system uses a lightweight Random Forest model, making it more suitable for embedded applications while still achieving high accuracy.

5.1.4 Discussion of Results

The system achieved high accuracy under controlled indoor conditions. The results confirm that CSI data can effectively capture variations caused by human movement. The feature extraction and classification process successfully distinguishes between walking, sitting, and standing activities.

However, the system was tested in a single environment with one user. In real-world scenarios involving multiple users or different environments, the performance may vary. Further improvements can be made to enhance robustness and scalability.

The proposed system successfully recognizes human activities using WiFi CSI data with high accuracy. It is a low-cost, portable, and real-time solution, making it suitable for practical applications in smart environments.

5.2 Conclusion

This project successfully demonstrates a WiFi CSI-based Human Activity Recognition system using the ESP32 microcontroller. The system operates in Access Point (AP) mode without requiring external WiFi infrastructure and captures CSI data across multiple subcarriers. The captured data is processed using statistical feature extraction techniques and classified using a Random Forest machine learning algorithm. The system provides real-time output through an OLED display and LED indicators. The hardware design, including the battery-powered circuit and output interface, makes the system portable and standalone. The use of a lightweight machine learning model ensures efficient performance suitable for embedded applications.

5.2 Future Scope

1. On-device TinyML inference: Deploy a quantised Random Forest or lightweight neural network directly on the ESP32 using TensorFlow Lite for Microcontrollers, eliminating laptop dependency completely.
2. External antenna upgrade: Implement the U.FL pigtail+2.4GHz dipole antenna modification to improve CSI signal quality, with documented RSSI improvement measurements.
3. Multi-subject HAR: Extend the system to recognise activities of multiple simultaneous subjects using multi-antenna CSI correlation techniques.
4. Additional activity classes: Expand the activity vocabulary to include falling, jumping, exercising, and other clinically or practically relevant activities
5. Cross-environment generalisation: Apply domain adaptation techniques to maintain classification accuracy across different room layouts and furniture configurations.
6. PCB design: Fabricate a custom PCB integrating the ESP32 module, power circuit, OLED, and LED indicators into a compact form factor suitable for permanent installation.

Bill of material

Sr.No	Component	Specification	Quantity	Cost
1.	ESP32 DevKit V1	WiFi + CSI Support	1	400
2.	18650 Li-ion Battery	3.7V, 2000mAh	1	100
3.	TP4056 Module	Battery Charging Module	1	50
4.	AMS1117 Voltage Regulator	3.3V Regulator	1	30
5.	OLED Display	0.96 inch, I2C (SSD1306)	1	150
6.	LEDs	Red, Yellow, Green	3	30
7.	Resistors	330Ω	3	10
8.	SPDT Switch	ON/OFF Switch	1	30
9.	Perfboard	General Purpose PCB	1	50
10.	Jumper Wires	Male/Female	1 set	50
11.	18650 Battery Holder	Single Cell Holder	1	40
Total				940

References

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